

PointSAT checkpoint scaling study — DRAFT

Protocol and interpretation

Five geometric families; the registered number of diverse SAT-only targets per family; one registered native seed per target. Every arm/budget starts afresh from the same frozen target and seed. Native allowances are 10, 30, 60 seconds, not whole-trial wall or CPU budgets. This is a small, deliberately selected, non-IID corpus, not a population success-rate estimate.

The primary endpoint is exact geometric validity of the final saved checkpoint. Any saved valid point set is secondary. For 19 points, printed-coordinate geometric validity is distinct from exact threefold-symmetry reconstruction and from realizing the initial SAT target. For the cap problem, caps are counted in the actual coordinate x direction.

The 19-point arms compare upstream continuous search, v4 continuous search, and v4 split-budget target repair plus warm restart. The last is a compound strategy, not an isolated feedback effect. Paper-family arms use upstream, v6 plain, v6 line, v6 line+pair, and v6 feedback. No time of actual first success is observed or interpolated. Solve-rate differences are not fixed-work kernel speedups.

Incomplete snapshot. Pending targets are not failures; hollow plot points are incomplete. Missing measurements remain unknown. SQL budget-effects and saved-summary snapshots can have slightly different cutoff times while workers run.

The nineteen corpus contains 9 fresh, 11 historical initial sat. All committed fresh targets are considered first, followed by historical initial SAT assignments in frozen source/line order. No coordinate-success or feedback-outcome selection is used. Historical proof-filtered targets are a reused stratum, not all-fresh generation. The known positive-control canonical type is excluded; retained targets satisfy the original CNF and the registered canonical/distance checks. Benchmark checks use the original CNF, not the extra generation cuts.

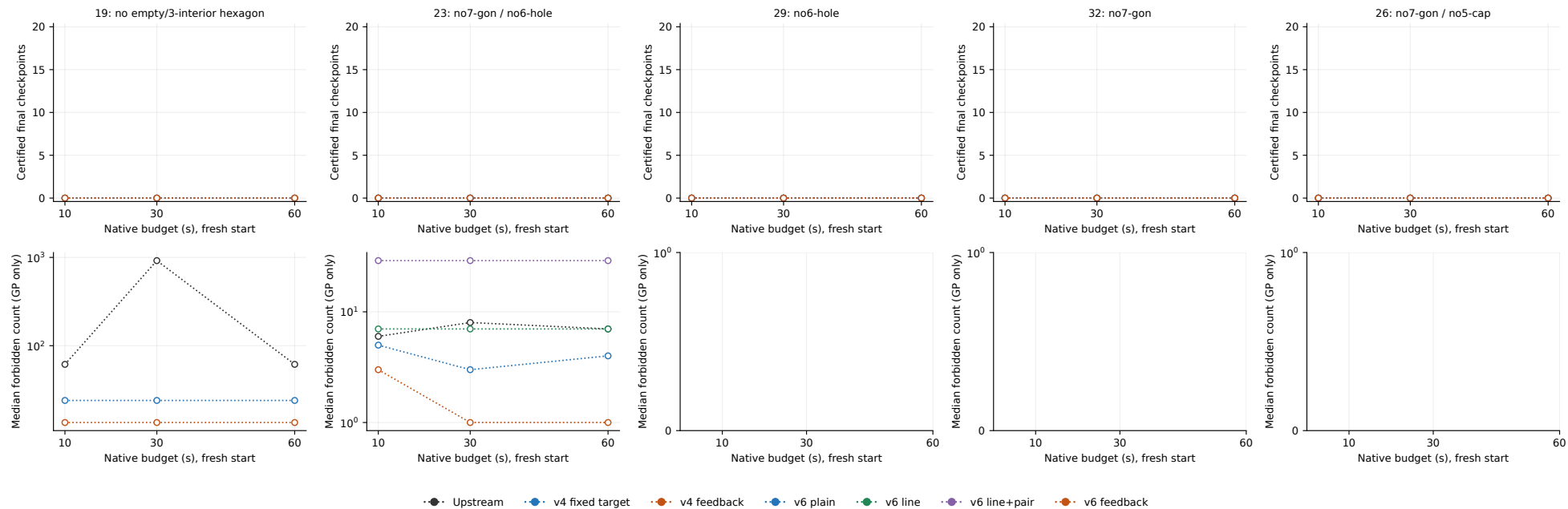
Final-checkpoint success counts

Each cell is valid final outputs / completed trials. The planned count is listed per problem/arm. Missing entries are not measured zeroes.

Problem	Arm	Planned	10 s	30 s	60 s
symmetry19	Upstream	20	0 / 12	0 / 14	0 / 12
symmetry19	v4 fixed target	20	0 / 12	0 / 12	0 / 12
symmetry19	v4 feedback	20	0 / 12	0 / 12	0 / 12
mixed23	Upstream	20	0 / 1	0 / 1	0 / 1
mixed23	v6 plain	20	0 / 1	0 / 1	0 / 1
mixed23	v6 line	20	0 / 1	0 / 1	0 / 1
mixed23	v6 line+pair	20	0 / 1	0 / 1	0 / 1
mixed23	v6 feedback	20	0 / 1	0 / 1	0 / 1
holes29	Upstream	20	0 / 0	0 / 0	0 / 0
holes29	v6 plain	20	0 / 0	0 / 0	0 / 0
holes29	v6 line	20	0 / 0	0 / 0	0 / 0
holes29	v6 line+pair	20	0 / 0	0 / 0	0 / 0
holes29	v6 feedback	20	0 / 0	0 / 0	0 / 0
gons32	Upstream	20	0 / 0	0 / 0	0 / 0
gons32	v6 plain	20	0 / 0	0 / 0	0 / 0
gons32	v6 line	20	0 / 0	0 / 0	0 / 0
gons32	v6 line+pair	20	0 / 0	0 / 0	0 / 0
gons32	v6 feedback	20	0 / 0	0 / 0	0 / 0
caps26	Upstream	20	0 / 0	0 / 0	0 / 0
caps26	v6 plain	20	0 / 0	0 / 0	0 / 0
caps26	v6 line	20	0 / 0	0 / 0	0 / 0
caps26	v6 line+pair	20	0 / 0	0 / 0	0 / 0
caps26	v6 feedback	20	0 / 0	0 / 0	0 / 0

Checkpoint scaling and actual residuals

DRAFT — Final-checkpoint scaling; small non-IID target corpus, one seed per target



Top: valid final checkpoints. Bottom: median actual forbidden polygon/cap count among general-position (GP) finals only. GP denominators and target-orientation errors are in README.md and report.json. Scales differ between families. Dotted connections describe independent budget starts, not continuous survival curves. Coincident arm values can overlap.

Paired effect of more native time

Gains/losses compare validity of the same target and seed. The residual delta is higher-budget minus lower-budget forbidden count, then a median across paired GP targets. It is not the difference of group medians. Both endpoints need a hash/content-checked stored audit certificate for inclusion in residual changes. Negative deltas are improvements; longer runs can lose final-checkpoint validity.

Problem	Arm	Pairs	10→30 G/L	Delta (GP)	Pairs	30→60 G/L	Delta (GP)
symmetry19	Upstream	12	0/0	0.00 (12)	12	0/0	0.00 (12)
symmetry19	v4 fixed target	12	0/0	0.00 (12)	12	0/0	0.00 (12)
symmetry19	v4 feedback	12	0/0	0.00 (12)	12	0/0	0.00 (12)
caps26	Upstream	0	0/0	− (0)	0	0/0	− (0)
caps26	v6 plain	0	0/0	− (0)	0	0/0	− (0)
caps26	v6 line	0	0/0	− (0)	0	0/0	− (0)
caps26	v6 line+pair	0	0/0	− (0)	0	0/0	− (0)
caps26	v6 feedback	0	0/0	− (0)	0	0/0	− (0)
gons32	Upstream	0	0/0	− (0)	0	0/0	− (0)
gons32	v6 plain	0	0/0	− (0)	0	0/0	− (0)
gons32	v6 line	0	0/0	− (0)	0	0/0	− (0)
gons32	v6 line+pair	0	0/0	− (0)	0	0/0	− (0)
gons32	v6 feedback	0	0/0	− (0)	0	0/0	− (0)
holes29	Upstream	0	0/0	− (0)	0	0/0	− (0)
holes29	v6 plain	0	0/0	− (0)	0	0/0	− (0)
holes29	v6 line	0	0/0	− (0)	0	0/0	− (0)
holes29	v6 line+pair	0	0/0	− (0)	0	0/0	− (0)
holes29	v6 feedback	0	0/0	− (0)	0	0/0	− (0)
mixed23	Upstream	1	0/0	2 (1)	1	0/0	-1 (1)
mixed23	v6 plain	1	0/0	-2 (1)	1	0/0	1 (1)
mixed23	v6 line	1	0/0	0 (1)	1	0/0	0 (1)
mixed23	v6 line+pair	1	0/0	0 (1)	1	0/0	0 (1)
mixed23	v6 feedback	1	0/0	-2 (1)	1	0/0	0 (1)

All registered budget comparisons, fewer/same/more counts, missing pair sides, non-GP/audit exclusions, and secondary orientation-error changes are preserved in report.json and the Markdown report.

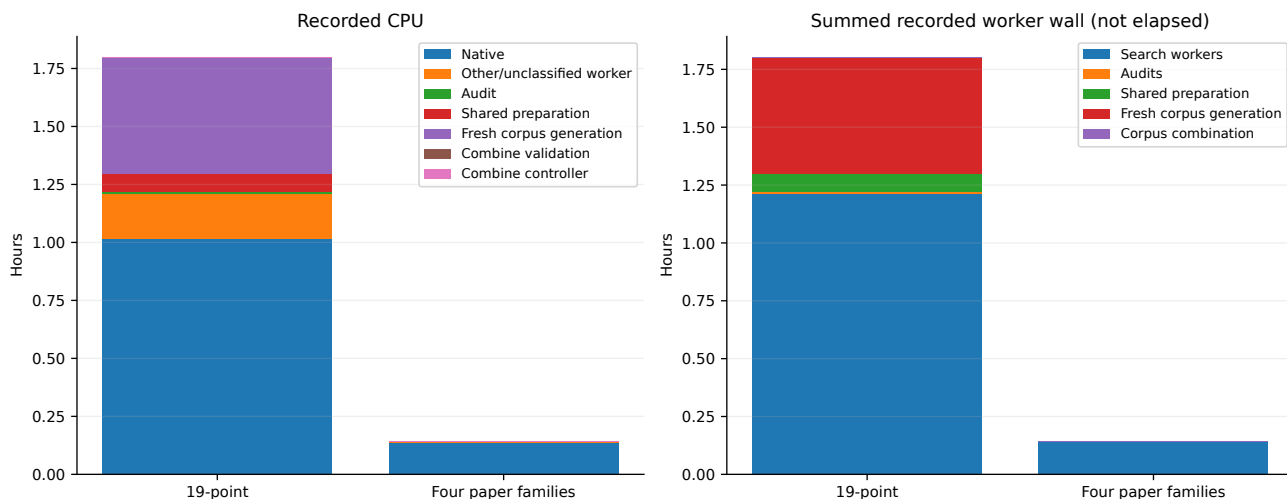
Compute accounting

CPU seconds measure processor consumption; summed worker-wall seconds are not elapsed calendar time. Native allowance excludes SAT feedback, initial flippability/preparation, and exact audits. Worker CPU beyond native includes initialization and unclassified work; it is not a pure SAT timer.

Study	Component	CPU seconds	Worker wall seconds
nineteen	Native	3652.73	3660.57
nineteen	Other/unclassified search worker	706.44	–
nineteen	Exact audits	32.40	34.59
nineteen	Shared preparation	268.83	270.09
nineteen	Fresh corpus generation	1798.74	1800.07
nineteen	Combined-corpus validation	12.54	14.38
nineteen	Combine controller	1.50	–
paper	Native	498.32	500.05
paper	Other/unclassified search worker	5.14	–
paper	Exact audits	4.93	5.24
paper	Shared preparation	1.13	1.15
paper	Fresh corpus generation	–	–
paper	Combined-corpus validation	–	–
paper	Combine controller	–	–

Shared preparation and corpus costs are counted once, not per arm/budget. Fresh-generation CPU and combined-corpus revalidation are separate. Historical generation sunk costs are not attributed to this batch. Combine wall includes its controller; do not add it again to child wall. Unknown CPU stays unknown; wall time is never substituted. Completed-summary costs and all-attempt totals are separate views and must not be added together.

DRAFT — Known cost components only; missing components are not measured zeros



Errors and censoring

nineteen: summary complete=False; persisted attempts=112, incomplete attempts=2, external interruptions=2, overhead watchdogs=0, audit watchdogs=0, forced native kills=0.

Recorded statuses: {"FINISHED": 110, "INTERRUPTED": 2}.

paper: summary complete=False; persisted attempts=15, incomplete attempts=0, external interruptions=0, overhead watchdogs=0, audit watchdogs=0, forced native kills=0.

Recorded statuses: {"FINISHED": 15}.

Ordinary native-budget termination is expected, not an operational error. Failed/missing certificates are not impossibility proofs. Interrupted/inflight work remains separately censored; abrupt termination may leave unrecorded CPU. The machine-readable ledger retains retries, errors, and cost-field coverage.

Separate positive control

This is one known realizable type, excluded from the diverse target corpus. Cold-start results are a pipeline sanity check, not a representative solve-rate estimate or part of the diverse-target denominators.

Arm	Geometry valid	Native CPU s	Native wall s	Target errors
Upstream	True	1.66	1.67	3
v4 fixed target	True	1.46	1.52	6

Provenance and reproduction

This PDF, README.md, report.json, scaling-report.tex, and the vector/PNG figures form one read-only report snapshot. The JSON records full input/output hashes, frozen corpus provenance, raw-attempt diagnostics, and paired effects. No searches or geometry audits are run during report generation. Earlier reports remain untouched.

nineteen registration: /home/andrew/PointSAT/benchmarks/compare19_scaling_20260907/registration.json

867cdb70c0622ab441ec830f92c2cb9eec7802738ca81306cf37a51faf420fe6

paper registration: /home/andrew/PointSAT/benchmarks/paper_scaling_20260907/registration.json

802ca2d1f810f6b6acf39bf29e1840a91db6c3672d056986ce92471502d4580c

Machine/environment snapshot: /home/andrew/PointSAT/benchmarks/scaling_machine_20260907.json. This does not imply exclusive ownership of every machine core.

Positive-control source: /home/andrew/PointSAT/benchmarks/compare19_positive_control_20260907/result.json